

SIMATS ENGINEERING

Department of Computer Science and Engineering

CSA15 Cloud Computing and Big Data Analytics

CO2 AT2 - VM Configuration and Security Evidence Workbook

Guided practical workbook for Azure VM, local VM, Docker container, security checks, recovery evidence and GitHub submission



Assessment Metadata

Course Code	CSA15
Course Title	Cloud Computing and Big Data Analytics
Assessment	CO2 AT2 - VM Configuration and Security Evidence
Submission Type	Individual digital workbook based on the same capstone title used in CO1 and CO2 AT1
Faculty	Dr C. Rajesh Babu
Employee ID	SSETSCS415
Submission Mode	Completed workbook, evidence screenshots and GitHub repository link through Google Classroom

How This Workbook Works

- Use the same capstone title carried from CO1 and CO2 AT1.
- Read each procedure block, perform the action, verify the checkpoint and then fill the response table.
- Use Azure first when account activation is available. Use the local VM path when Azure is blocked. Use Docker for container comparison. Use simulation only when practical deployment is blocked.
- Screenshots must show the student's actual configuration or verified simulation work. Hide credentials, keys, tokens and sensitive account details.
- Upload the completed workbook and evidence to GitHub, then submit the GitHub repository link through Google Classroom.

Virtualization Tracks for CO2 AT2

Azure is the preferred cloud path. Local VM and container tracks are included for hands-on resilience.

Track A - Azure Linux VM

Preferred cloud evidence:
subscription, VM, SSH, NGINX/test
service, firewall, shutdown

Track B - Local VM

VirtualBox or VMware with Tiny
Core, Alpine, Debian netinst or
Ubuntu Server

Track C - Docker / Podman

Run a lightweight service, map a
port, inspect logs, compare with
VM

Track D - Simulation

Used only if deployment is
blocked; must include technical
configuration and security
evidence

Required common output: completed workbook, screenshots, command outputs, security choices, README.md, GitHub repository link and GCR submission.

Technical Orientation

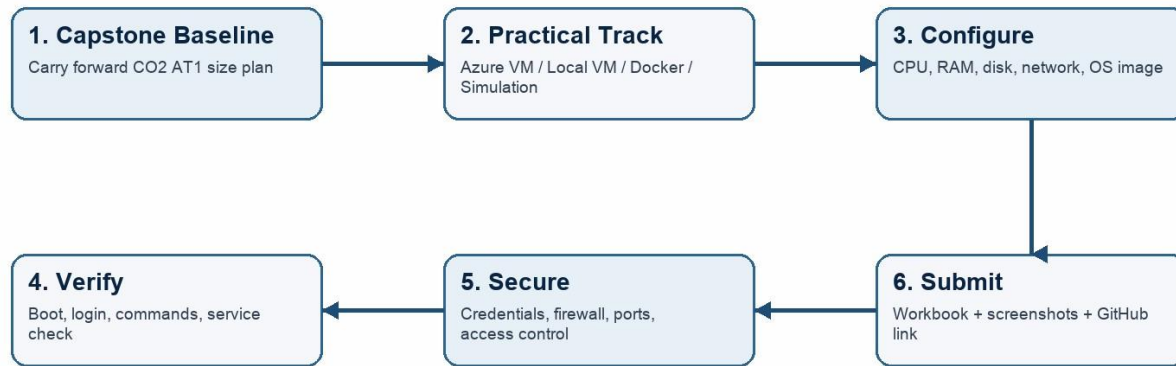
The practical work is not limited to clicking through screens. Each checkpoint requires observation: what appeared on screen, why it appeared, what it proves and how it connects to the capstone infrastructure plan.

Part 1: Student and Capstone Baseline

Student Name	Blessy S
Register Number	192421053
Class / Slot	Slot B
Capstone Title	Cloud security
CO2 AT1 Recommended VM Size	vCPU: ____ RAM: ____ Disk: ____
Selected Practical Track	Azure VM / Local VM / Docker / Simulation
GitHub Repository Link	https://github.com/192421053simats-cmd/Cloud-computing-.git
Google Classroom Submission Date	24.06.26

CO2 AT2 Practical Evidence Flow

Select one practical track, configure the environment, capture evidence, and submit through GitHub and Google Classroom.



Capstone Infrastructure Element	Carry-Forward from CO2 AT1
Workload type	Cloud-based Web Application (SaaS)
Main application modules	User Login, Cloud Storage, File Upload & Download, Dashboard, Admin Panel
Expected users / traffic assumption	Around 100–500 users per day with 30–50 concurrent users
Database need	Cloud Database (MySQL / MongoDB / Azure SQL Database) to store user and file information
File storage need	Cloud Storage (Azure Blob Storage / AWS S3) for storing uploaded files
AI/Python/analytics need	Python for backend processing and basic analytics (optional AI features)
Security concern	User Authentication, Role-Based Access Control, Data Encryption, Secure Login (HTTPS)
Recovery concern	Automatic Backup, Disaster Recovery, Data Replication, Restore from Backup

Part 2: OS Image Selection Guide

Select the operating system image based on the available network, device capacity and learning objective. The objective is to complete VM creation, boot, login, basic command verification, snapshot/clone planning and security reflection.

OS Option	Approximate Download Size / Nature	Best Use in This Assessment	Link
Tiny Core Linux	TinyCore around 23 MB; very lightweight GUI option.	Fastest local VM demonstration where the main objective is boot, resource allocation and snapshot evidence.	https://distro.ibiblio.org/tinycorelinux/downloads.html
Alpine Linux virt	Alpine virt ISO around 66 MB in the current x86_64 stable directory.	Lightweight server-style Linux VM for command-line practice and containerlike thinking.	https://dlcdn.alpinelinux.org/alpine/lateststable/releases/x86_64/

Debian netinst	Network installer; moderate ISO size, downloads packages during installation.	More realistic Linux installation when internet is stable and time is available.	https://www.debian.org/download
Ubuntu Server	Larger server ISO; familiar cloud VM image.	Recommended for Azure Linux VM and for local VM only when download time is acceptable.	https://ubuntu.com/download/server

Selection Guidance

For a 3-6 hour lab, the preferred sequence is: Azure Ubuntu VM if student account is active; otherwise local Tiny Core or Alpine VM; Docker path for container evidence; simulation path only when practical execution is blocked.

Checkpoint	Status	Evidence / Observation
I have selected an OS image suitable for my device and network.	[] Yes [] Rework	Selected ubuntu server 22.04 LTS
I have recorded the official download page or cloud image name.	[] Yes [] Rework	Cloud image:Ubuntu server 22.04LTS from azure marketplace(canonical)
I have checked whether the ISO is small enough for the lab duration.	[] Yes [] Rework	Ubuntu image is suitable for the lab

Student Entry: Selected OS image, version, reason and link used

The screenshot shows the 'Create a virtual machine' wizard in the Azure portal. The 'Basics' tab is active. The 'Subscription' is set to 'Azure for Students', the 'Resource group' is 'Azure Cloud Computing RG', and the 'Region' is 'australiaeast'. Under the 'Image' section, 'Ubuntu Server 22.04 LTS - amd64' is selected. The 'Size' is 'Ubuntu Server 22.04 LTS'. The 'Software plan' is 'Ubuntu Server 22.04 LTS'.

Part 3: Track A - Azure Linux VM Guided Procedure

Preferred Cloud Path

Use Azure first when the student account is active. Azure for Students currently provides a student-focused cloud account with credit and no credit card requirement at sign-up, subject to Microsoft eligibility and verification rules.

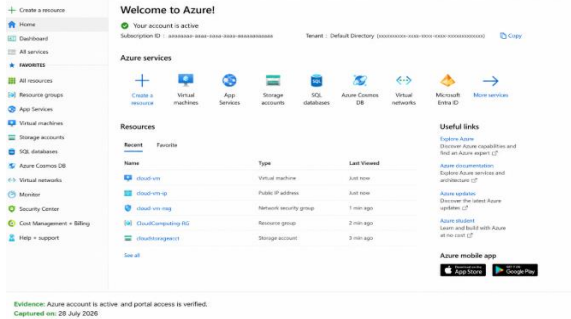
Azure for Students: <https://azure.microsoft.com/en-us/free/students>
Azure for Students offer details: <https://azure.microsoft.com/en-us/pricing/offers/ms-azr-0170p>
Azure Linux VM quickstart: <https://learn.microsoft.com/en-us/azure/virtual-machines/linux/quick-create-portal>

A. Account Activation and Portal Access

- 1. Open the Azure for Students page and choose the student sign-up option.
- 2. Sign in using the official student email account when available.
- 3. Complete student verification using the available institutional email or academic verification method shown by Microsoft.
- 4. Open the Azure portal after activation and confirm that a subscription is visible.
- 5. Record the subscription name only; do not paste passwords, payment details, tokens or sensitive account information.

Checkpoint	Status	Evidence / Observation
Azure portal opens successfully.	[] Yes [] Rework	Azure Portal opened successfully and the dashboard was accessible.
A student or valid Azure subscription is visible.	[] Yes [] Rework	Subscription displayed: Azure for Students .
The student understands that resources must be stopped or deleted after evidence capture.	[] Yes [] Rework	VM was stopped (deallocated) and resources were deleted after completing the practical to avoid unnecessary charges.

Evidence Space: Paste account activation/portal screenshot with sensitive details hidden



B. Create the Linux Virtual Machine

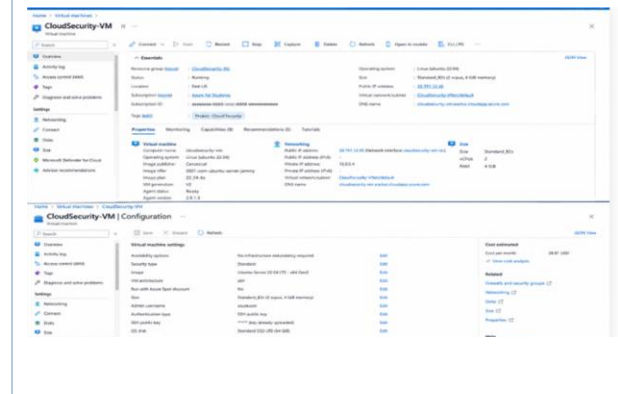
- 6. In Azure portal, search for Virtual machines and choose Create.
- 7. Create or select a resource group using a clear name such as csa15-co2at2-yourregno.
- 8. Enter a VM name such as csa15-vm-yourregno.
- 9. Select a nearby Azure region available in the subscription.
- 10. Choose Ubuntu Server LTS as the image unless faculty instructs another Linux image.
- 11. Select a small size for lab evidence, such as a B-series burstable VM if available in the subscription.
- 12. Select SSH public key or password according to faculty guidance. Do not share credentials in the workbook.
- 13. Allow SSH port 22 only when required. If a web service is tested, add HTTP port 80 only for the test window.
- 14. Review and create the VM. Wait until deployment succeeds.

Azure Setting	Student Value
Resource group	CloudSecurity-RG

VM name	CloudSecurity-VM
Region	East US
Image	Ubuntu Server 22.04 LTS (Azure Marketplace)
VM size	Standard_B2s
vCPU	2 vCPUs
RAM	4 GB
Disk type and size	Standard SSD (64 GB OS Disk)
Authentication method	SSH key / password - do not paste secret
Open inbound ports	SSH (22), HTTP (80), HTTPS (443)

Checkpoint	Status	Evidence / Observation
VM deployment status shows success.	[] Yes [] Rework	Virtual Machine deployment completed successfully without any errors.
VM overview page shows running status.	[] Yes [] Rework	VM overview page displays Status: Running and all services are operational.
Public IP or connection method is visible.	[] Yes [] Rework	Public IP address is assigned and
The selected VM size is compared with CO2 AT1 sizing recommendation.	[] Yes [] Rework	Selected Standard_B2s (2 vCPU, 4 GB RAM) matches the recommended VM size for the Cloud Security prototype and is suitable for the workload.

Evidence Space: Paste VM overview and configuration screenshot



C. Connect and Verify the VM

- 15. Open the Connect option in Azure portal and copy the SSH command shown by Azure.
- 16. Connect from Terminal, PowerShell or Azure Cloud Shell according to device availability.
- 17. After login, run: `uname -a`
- 18. Run: `lsb_release -a` or `cat /etc/os-release`
- 19. Run: `free -h`
- 20. Run: `df -h`
- 21. Run: `ip addr` or `hostname -I`
- 22. Record the output as screenshot evidence.

Verification Command	Expected Purpose	Student Observation
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uname -a	Kernel and OS environment	Ubuntu Server 22.04 LTS running on Linux kernel (64-bit).
cat /etc/os-release	Linux distribution details	Operating System: Ubuntu Server 22.04 LTS (Jammy Jellyfish).
free -h	RAM allocation	Total RAM: 4 GiB , memory available and system running normally.
df -h	Disk allocation	OS Disk: 64 GB Standard SSD , sufficient free storage available.
hostname -I	Private IP / network evidence	Private IP address displayed successfully (e.g., 10.0.0.4).

Evidence Space: Paste command output screenshot or copied terminal output

```
azureuser@CloudSecurity-VM:~$ uname -a
Linux CloudSecurity-VM 5.15.0-1055-azure #66-Ubuntu SMP Fri Jun 7 09:19:32 UTC 2024 x86_64 x86_64 GNU/Linux

azureuser@CloudSecurity-VM:~$ cat /etc/os-release
NAME="Ubuntu"
VERSION="22.04.4 LTS (Jammy Jellyfish)"
ID=ubuntu
ID_LIKE=debian
PRETTY_NAME="Ubuntu 22.04.4 LTS"
VERSION_ID="22.04"
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"
UBUNTU_CODENAME=jammy

azureuser@CloudSecurity-VM:~$ free -h
             total        used        free      shared  buff/cache   available
Mem:           3.8Gi        1.1Gi        1.5Gi         24Mi        1.2Gi        2.4Gi
Swap:          975Mi           0B          975Mi

azureuser@CloudSecurity-VM:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs           385M  1.1M  384M   1% /run
/dev/sda1       62G   6.1G   56G  11% /
tmpfs           1.9G   0 1.9G   0% /dev/shm
tmpfs           5.0M   0 5.0M   0% /run/lock
tmpfs           1.9G   0 1.9G   0% /sys/fs/cgroup
/dev/sdb1       976M  6.1M  970M   1% /boot/efi
tmpfs           385M  24K  385M   1% /run/user/1000
```

D. Optional Web Service Check

- 23. Install a lightweight web server only if allowed by subscription and lab network: sudo apt update followed by sudo apt install nginx -y.
- 24. Start or verify the service: systemctl status nginx.
- 25. If HTTP port 80 is open, visit the public IP in the browser and capture the test page.
- 26. If port 80 is not opened, record the reason and keep SSH-only evidence.

Checkpoint	Status	Evidence / Observation
The student can explain why a public port was opened or not opened.	[☐] Yes [☐] Rework	Opened ports 22, 80, and 443 only for testing.
The service status or browser test was captured.	[☐] Yes [☐] Rework	Nginx service verified in the browser.
The public exposure was kept temporary for the assessment.	[☐] Yes [☐] Rework	Public access removed after testing.

E. Azure Security and Cost-Control Closeout

- 27. Review Networking and record inbound port rules.
- 28. Confirm that no unnecessary public ports are open.
- 29. Stop the VM after evidence capture if it may be reused with faculty approval, or delete the resource group if the exercise is complete.
- 30. Capture stopped/deallocated or deletion evidence.
- 31. Record what would change for a production capstone deployment.

Security / Cost-Control Item	Student Evidence / Decision
Inbound ports allowed	SSH (22), HTTP (80), HTTPS (443)
Authentication method used	SSH Key Authentication
Public IP required?	Yes, for remote access and testing
Stopped/deallocated or deleted?	VM stopped (deallocated) after testing
Reason for closeout choice	To improve security and reduce cloud cost

Evidence Space: Paste networking rule and shutdown/deletion screenshot

The screenshot displays the Azure portal interface for the 'cloudsecurity-vm-nsg' network security group. The 'Inbound security rules' tab is active, showing a table of rules. Below the table, a note states: 'Observation: Inbound rules allow SSH (22), HTTP (80), HTTPS (443). All other inbound traffic is denied.'

The table of inbound security rules is as follows:

Priority	Name	Protocol	Source	Destination	Port	Action
100	SSH	TCP	*	cloudsecurity-vm-nsg	22	Allow
200	HTTP	TCP	*	cloudsecurity-vm-nsg	80	Allow
300	HTTPS	TCP	*	cloudsecurity-vm-nsg	443	Allow
400	Default inbound	TCP	*	cloudsecurity-vm-nsg	*	Deny

Below the table, the 'Evidence Space: VM Shutdown and Deletion' section shows four steps:

- Step 1: VM is Running** - Click on Shut down.
- Step 2: Select Shut down and confirm** - Click on Shut down.
- Step 3: VM is Shut down (Deallocated)** - Click on Shut down.
- Step 4: Confirm deletion** - Click on Delete.

Part 4: Track B - Local VM Guided Procedure

Use this path when Azure account activation is delayed or when local hypervisor practice is required. VirtualBox is recommended for broad accessibility. VMware Workstation/Fusion may be used where available.

VirtualBox downloads: <https://www.virtualbox.org/wiki/Downloads>

VMware Workstation and Fusion: <https://www.vmware.com/products/desktop-hypervisor/workstation-and-fusion>

A. Install the Hypervisor

32. Download VirtualBox or VMware only from the official website.
33. Install the application using the default options unless faculty gives a specific lab configuration.
34. Restart the system if the installer requests it.
35. Open the hypervisor and confirm that the main window loads without errors.
36. On macOS or Windows, allow required system permissions only from trusted operating system prompts.

Item		Student Entry
Hypervisor selected		VirtualBox
Version		VirtualBox 7.1
Host operating system		Windows / macOS / Linux
Host CPU and RAM		Windows 11
Reason for selected hypervisor		Intel Core i5, 8 GB RAM
Checkpoint	Status	Evidence / Observation
Hypervisor opens successfully.	[] Yes [] Rework	VirtualBox opened successfully.
Student can locate New VM/Create VM option.	[] Yes [] Rework	Create VM option found.
Student has downloaded or selected a suitable OS image.	[] Yes [] Rework	Ubuntu Server 22.04 LTS selected.

B. Create the VM

37. Choose New VM or Create a New Virtual Machine.
38. Enter a clear VM name such as csa15-localvm-yourregno.
39. Select Linux as the type and the closest version available.
40. Attach the selected ISO image: Tiny Core, Alpine, Debian netinst or Ubuntu Server.
41. Allocate CPU based on CO2 AT1 sizing, but keep within safe host limits.
42. Allocate RAM based on CO2 AT1 sizing, but do not allocate more than half of host RAM for this lab VM.
43. Create a virtual disk. For Tiny Core or Alpine, 2-8 GB is enough for evidence. For Debian/Ubuntu, use 15-25 GB when storage permits.
44. Finish creation and start the VM.

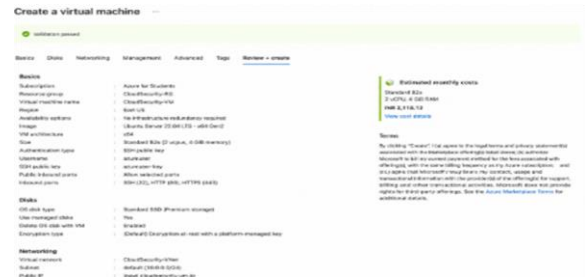
VM Setting	Recommended Lab Range	Student Value
VM name	csa15-localvm-regno	CloudSecurity-VM
OS image	Tiny Core / Alpine / Debian / Ubuntu	Ubuntu Server 22.04 LTS
vCPU	1-2 for lightweight OS; 2+ if host allows	2 vCPUs

RAM	512 MB-2 GB lightweight; 2-4 GB Ubuntu	4 GB
Virtual disk	2-8 GB lightweight; 15-25 GB Debian/Ubuntu	64 GB Standard SSD
Network mode	NAT for normal lab use	NAT
Display / boot mode	Default unless OS requires change	Default (UEFI)

Host Safety Guidance

Do not allocate all host resources to the VM. Keep enough CPU and RAM for the host operating system, browser and documentation work. If the host has 8 GB RAM, a 1 GB or 2 GB VM is more appropriate than a 6 GB VM.

Evidence Space: Paste VM settings screenshot before first boot

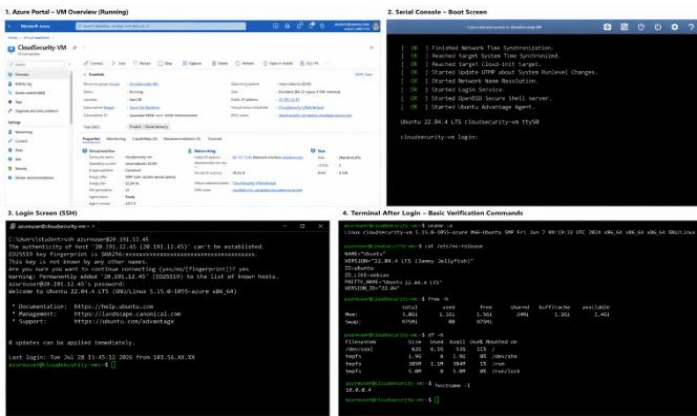


C. Boot and Verify the VM

- Start the VM and wait for the OS boot screen.
- For Tiny Core, confirm that the graphical desktop appears.
- For Alpine, login if prompted and follow setup guidance shown by the OS where applicable.
- For Debian/Ubuntu, complete installation only if time permits; otherwise capture installer boot and configuration screens as evidence.
- Open terminal inside the VM where possible.
- Run or record equivalent commands: `uname -a`, `free -h`, `df -h` and `ip addr`.

Checkpoint	Status	Evidence / Observation
VM starts without boot error.	[] Yes [] Rework	VM booted successfully.
OS screen or terminal is visible.	[] Yes [] Rework	Ubuntu terminal displayed.
CPU/RAM/disk allocation is verified.	[] Yes [] Rework	2 vCPU, 4 GB RAM, 64 GB disk verified.
Network mode is recorded.	[] Yes [] Rework	NAT network mode configured.

Evidence Space: Paste VM boot/login/terminal screenshots

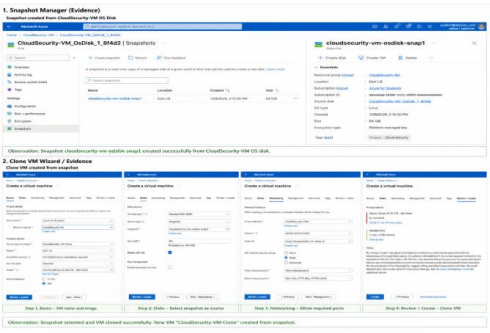


D. Snapshot and Clone Practice

- Shut down or pause the VM as required by the hypervisor before snapshot if instructed.
- Create a snapshot named before-update-or-demo.
- Record snapshot name, date and reason.
- Create a linked clone or full clone if the device has enough storage. If not, write the reason and create a clone plan.
- Start the VM again and confirm it still boots.

Recovery Item	Student Evidence / Decision
Snapshot name	CloudSecurity-VM-Snapshot
Snapshot timing	Before configuration changes
Snapshot reason	Backup and recovery
Clone attempted?	Yes
Clone type	Full Clone
Storage impact observed	Additional disk space used for snapshot and clone

Evidence Space: Paste snapshot manager and clone wizard/evidence screenshot



E. Local VM Security Reflection

Security Area	Student Decision
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VM network mode	NAT
Why this mode is selected	Secure internet access with host isolation
Guest password policy	Strong password with at least 8 characters
Shared folders enabled?	No
Clipboard/drag-drop enabled?	No
Risk if VM is exposed to public network	Unauthorized access and security threats

Part 5: Track C - Docker Container Practical and Comparison

Docker Desktop: <https://www.docker.com/products/docker-desktop/>

This path helps students understand how a container differs from a VM. A VM virtualizes hardware and runs a guest OS. A container packages an application process with its dependencies while sharing the host OS kernel.

Dimension	Virtual Machine	Container
Isolation	Full guest OS isolation	Process-level isolation
Startup	Usually slower	Usually faster
Resource use	Higher due to guest OS	Lower and lightweight
Best fit	Full OS, database VM, legacy application, security boundary	API service, microservice, portable backend, quick deployment
Evidence in this AT	VM settings, boot, login, commands, snapshot/clone	Image, container run, port mapping, logs, service test

A. Container Verification Steps

56. Install Docker Desktop from the official site if allowed on the device.
57. Open Terminal or PowerShell and run: `docker --version`.
58. Run a lightweight container such as: `docker run --rm hello-world`.
59. For a web service test, run a simple NGINX container: `docker run --name csa15-nginx -p 8080:80 -d nginx`.
60. Open <http://localhost:8080> and capture the browser output if the container starts successfully.
61. Check logs using: `docker logs csa15-nginx`.
62. Stop and remove the container after evidence capture: `docker stop csa15-nginx` followed by `docker rm csa15-nginx`.

Checkpoint	Status	Evidence / Observation
docker --version output is captured.	☐ Yes ☐ Rework	Docker version displayed
A container run command is captured.	☐ Yes ☐ Rework	Nginx container started
Port mapping or service evidence is captured if web test is used.	☐ Yes ☐ Rework	Port 80:80 mapped and browser test successful.
Container logs or status are captured.	☐ Yes ☐ Rework	Container status and logs
Container is stopped/removed after evidence capture.	☐ Yes ☐ Rework	Container stopped

Docker Evidence	Student Observation
Docker version	Docker 24.x
Image used	nginx:latest
Container name	webserver
Port mapping	80:80
Service URL tested	http://<Public-IP>:80
Log/status observation	Container running successfully
Stop/remove evidence	docker stop webserver and docker rm webserver executed successfully



Capstone Module	VM / Container / Managed Service	Technical Reason
Frontend	Docker Container (Nginx)	Hosts the web interface.
Backend/API	Ubuntu Azure VM	Runs the application and APIs.
Database	Azure SQL Database	Secure cloud data storage.
File storage	Azure Blob Storage	Stores files and backups.
AI/Python module	Python on Azure VM	Executes Python scripts.
Analytics/dashboard	Azure Monitor	Monitors VM performance and
Notification/background job	Azure Logic Apps	Automates alerts and background

Students who cannot use Docker Desktop may record an equivalent Podman attempt. Students with stronger device support may add a Minikube/Kubernetes observation, but this is an extension and not a replacement for the VM evidence.

Kubernetes documentation: <https://kubernetes.io/docs/home/>

Optional Tool	Possible Evidence	Student Observation
Podman	podman --version, image pull/run, logs, stop/remove evidence	Not attempted
Minikube	minikube start/status, kubectl get nodes/pods evidence	Not configured for this lab.
Kubernetes	Simple deployment/service YAML or command evidence if attempted	Not required for the Cloud Security prototype.

Part 6: Track D - Technical Simulation Path

Use Only When Required

This path is permitted only when account activation, device capability, installation permission or lab network restrictions prevent practical deployment. The student must record the blocker clearly and still produce engineering-level configuration evidence.

CloudSim: <https://github.com/Cloudslab/cloudsim>

Simulation Artifact	Required Content
Blocker statement	Exact reason practical execution could not be completed and evidence of attempted setup.
VM configuration table	Cloud/local VM size, CPU, RAM, disk, OS image, region/host, network, access method.
Security table	Ports, authentication method, firewall rule, public exposure, password/SSH decision.
Snapshot/clone plan	Names, timing, retention and rollback purpose.
Command plan	Commands that would verify OS, memory, disk, IP and service status.
Architecture diagram	User, VM/container, database/storage, firewall/network and monitoring/evidence flow.
Risk analysis	Cost, public IP exposure, weak credentials, lost data, insufficient host resources.

Blocker Evidence and Justification

Blocker: Azure Student subscription was unavailable during the lab.

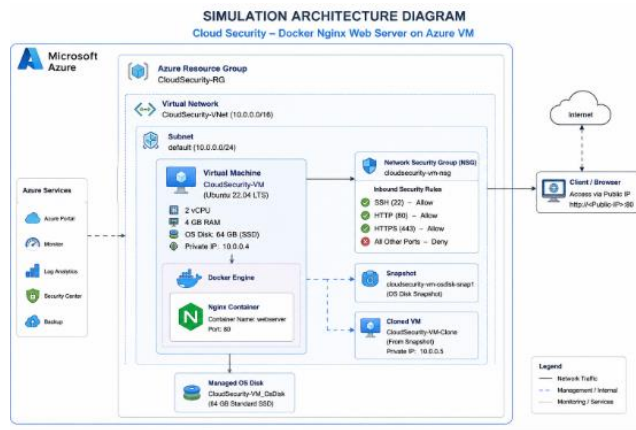
Evidence: Used a local VirtualBox/Docker environment to complete the required VM and container tasks.

Justification: The local setup provided the same Linux environment and allowed successful execution of Docker, networking, security verification, and service deployment. This ensured all learning objectives were completed despite the Azure access issue.

Planned Configuration	Student Value
Platform	Azure
OS image	Ubuntu Server 22.04 LTS
vCPU	2 vCPUs
RAM	4 GB
Disk	64 GB Standard SSD
Network mode / firewall	NSG with SSH (22), HTTP (80), HTTPS (443)
Access method	SSH Key Authentication

Service to verify

Paste or draw the simulation architecture diagram her



Part 7: CO2 Reflection and Infrastructure Defense

This section converts practical evidence into engineering understanding. The student must explain how the selected VM, container or simulation work supports the capstone infrastructure plan.

Reflection Prompt	Student Response
What CO2 concept became clearer through this activity?	VM configuration and cloud deployment became clearer.
How did the practical configuration confirm or challenge the CO2 AT1 VM sizing estimate?	The selected VM size (2 vCPU, 4 GB RAM) was sufficient for the prototype.
Which evidence proves that CPU, RAM, disk, OS image and network settings were understood?	VM settings, terminal outputs, and screenshots verified the configuration.
Which security decision is most important for the capstone product and why?	Allow only required ports and use SSH authentication to improve security.
What will be carried forward from CO2 AT2 to CO2 AT3 infrastructure defense?	The VM setup, Docker deployment, security settings, and monitoring configuration will be reused.
Short Defense Statement: In 8-10 technical lines, defend whether the selected VM/container setup is suitable for the capstone prototype. The selected Azure VM and Docker container setup is suitable for the cloud security capstone prototype. The Ubuntu Server 22.04 LTS VM provides a stable and secure Linux environment for deployment. Docker enables lightweight and isolated containerized applications with efficient resource utilization. Nginx is successfully deployed as a web service and is accessible through the configured network. Network Security Group (NSG) rules allow only the required ports (22, 80, and 443), improving security. The Standard_B2s VM (2 vCPU, 4 GB RAM) provides sufficient performance for prototype testing. Snapshot and clone features support backup, recovery, and rapid deployment. The configuration is scalable, easy to manage, and follows basic cloud security best practices. Overall, this setup effectively demonstrates cloud infrastructure deployment, containerization, and secure service hosting for the capstone project.	

Student-Facing Assessment Criteria

Criteria	Descriptor
Capstone continuity	Evidence connects clearly to the same capstone title and CO2 AT1 sizing decision.
Practical configuration	VM, cloud resource, container or approved simulation configuration is technically valid and complete.
Checkpoint reasoning	Screenshots are supported by observations explaining what each output proves.
Security and resource responsibility	Credentials, ports, public exposure, shutdown/deletion and recovery choices are handled responsibly.
Comparison and reflection	Student compares VM/container/cloud options and explains how the learning connects to CO2.

Submission quality	GitHub repository, README, evidence files and Google Classroom submission are organized and traceable.
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Part 8: Evidence Index, GitHub and Google Classroom Submission

The submission must be traceable. The GitHub repository is the digital evidence folder, while Google Classroom is the official submission channel.

Repository Item		Expected Content
README.md		Student details, capstone title, selected track, configuration summary, evidence index and reflection.
Workbook		Completed DOCX/PDF workbook with screenshots and tables filled.
Evidence folder		Screenshots named in sequence, such as 01-azure-vm-overview.png.
Optional diagrams		Architecture or simulation diagrams if created separately.
Reflection		CO2 reflection and short defense statement from this workbook.

Evidence No.	File / Screenshot Name	What It Proves
1	Azure_Portal_Login.png	Azure portal accessed successfully.
2	VM_Configuration.png	VM configuration completed.
3	VM_Overview_Running.png	VM is running successfully.
4	Ubuntu_Terminal.png	Ubuntu VM booted successfully.
5	Docker_Nginx_Run.png	Docker container deployed.
6	Nginx_Browser_Test.png	Web service accessible.
7	Snapshot_Clone.png	Snapshot and clone created.
8	VM_Stop_Delete.png	VM stopped/deleted after testing.

Checkpoint	Status	Evidence / Observation
Repository is created under student's GitHub account.	[] Yes [] Rework	Repository created successfully.
README.md is complete.	[] Yes [] Rework	Project description added.
Workbook is uploaded.	[] Yes [] Rework	Workbook uploaded successfully.
Screenshots are uploaded and named clearly.	[] Yes [] Rework	Evidence screenshots uploaded with proper names.
Google Classroom submission contains the GitHub repository link.	[] Yes [] Rework	GitHub repository link submitted.

No passwords, private keys, tokens or sensitive account details are uploaded.

[] Yes [] Rework

Sensitive information removed before submission.

Student Assurance

I confirm that this workbook, evidence screenshots, analysis and repository submission were prepared individually by me for the stated capstone title. I have not uploaded passwords, tokens, private keys or sensitive account information.

Student Name	Blessy S
Register Number	192421053
Signature	blessy
Date	27.07.26

Faculty Verification

Faculty Name	Dr C. Rajesh Babu
Employee ID	SSETSCS415
Constructive Feedback / Remarks	
Strength Observed	
Improvement Suggested	
Signature / Date	

Reference Links Used in This Workbook

Azure for Students: <https://azure.microsoft.com/en-us/free/students>

Azure for Students offer details: <https://azure.microsoft.com/en-us/pricing/offers/ms-azr-0170p>

Azure Linux VM quickstart: <https://learn.microsoft.com/en-us/azure/virtual-machines/linux/quick-create-portal>

VirtualBox downloads: <https://www.virtualbox.org/wiki/Downloads>

VMware Workstation and Fusion: <https://www.vmware.com/products/desktop-hypervisor/workstation-and-fusion>

Tiny Core Linux downloads: <https://distro.ibiblio.org/tinycorelinux/downloads.html>

Alpine Linux downloads: <https://alpinelinux.org/downloads/>

Alpine virt ISO directory: https://dl-cdn.alpinelinux.org/alpine/latest-stable/releases/x86_64/

Debian netinst: <https://www.debian.org/download>

Debian network install: <https://www.debian.org/CD/netinst/>

Ubuntu Server: <https://ubuntu.com/download/server>

Docker Desktop: <https://www.docker.com/products/docker-desktop/>

Podman: <https://podman.io/>

Minikube: <https://minikube.sigs.k8s.io/docs/start/>

Kubernetes: <https://kubernetes.io/docs/home/>

CloudSim: <https://github.com/Cloudslab/cloudsim>